	Hence, the net force on the block is <u>downward</u>, opposite to the displacement.
2a	$g = GM/R^2$, so if distances near surface of planet is much smaller than R, g is a constant. force on rocket = mg, where m is mass of rocket, (g is the acceleration of free fall / gravitational field strength) ΔE_P = force × distance moved (or work done) $= mgh$ OR $\Delta E_P = m\Delta\Phi$ where m is mass of rocket $= GMm(1/R - 1/(R+h))$ $= GMmh/(R(R+h))$ distances h near surface of planet is much smaller than R, hence $(R+h) \approx R$ and $g = GM/R^2$ $\Delta E_P = mgh$
2b	Increase in GPE = Decrease in KE Increase in GPE = $GMm\left(\frac{1}{2R}-\frac{1}{5R}\right) = 4.0 \times 10^{14} \times 4.7 \times 10^4 \times \left(\frac{1}{2R}-\frac{1}{5R}\right)$ Decrease in KE = $(1.17 - 0.35) \times 10^{12}$ J $3R = 2.0634 \times 10^7 = 2.1 \times 10^7$ m
2c	Similarity: Gravitational fields are directed towards the planet and the rocket. Difference (any one): - Gravitational field at surface of planet is at orders of magnitude greater than that at the surface of the rocket. - Gravitational field at surface of planet has radial symmetry or is directed radially, but that at the surface of the rocket is not.

	- Magnitude of gravitational field at surface of planet is uniform, but that at the surface of the rocket varies.
3a	period = 0.36 s acceleration = $\omega^2 x$